

LCP = [1, 0, 1, 0, 0, 0, 0]

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STACK, QUEUE, AND LINKED LIST QUESTIONS

ACTIVITY - 1

Stack and queue operation

push(54); push(52); pop(); push(~~55~~); push(62); s = pop()

[54, 52] [52] [54, 55] [54, 55, 62] [62]

s = 62

enqueue(21); enqueue(24); dequeue(); enqueue(28); enqueue(30)

q = dequeue();

[21, 24]

[21]

[24, 28]

[24, 28, 30]

[24]

q = 24

s + q = 62 + 24 = 86

2. The result evaluating the postfix expression,

$$10 \ 5 + 60 \ 6 / \times 8 -$$

$$\times \ 10 \ 5 + = 15$$

$$\times \ 60 \ 6 / = 10$$

$$\times \ 15 \times 10 = 150$$

$$\times \ 150 - 8 = 142 //$$

$$3, \ 8 \ 2 \ 3 \wedge / 2 \ 3 \times + 5 \ 1 \times -$$

Stack after the first * is evaluated,

Symbol	Stack
8	8
2	8, 2
3	8, 2, 3
*	8, 8 $(2 \times 3 = 6)$
/	1 $(8/8 = 1)$
2	1, 2
3	1, 2, 3
*	1, 6 $(2 \times 3 = 6)$

$\therefore$ The top two elements are, 1, 6

$$4, \ a + b \times c - d \ \text{leaf}$$

Character	Stack	O/p
a		a
+	+	a
b	+	ab
*	+*	ab
c	+*	abc
-	-	abc * +

d	-	abc * + d
a	- a	abc * + d
e	- A	abc * + de
n	- AA	abc * + de
f	- nn	abc * + def
Empty		abc * + def nn -

5. Stack operations.

operation	Stack	O/P
push(10)	10	-
push(20)	10, 20	-
pop()	10	20
push(10)	10, 10	-
push(20)	10, 10, 20	-
pop	10, 10	20
pop	10	10
pop	empty	10
push(20)	20	-
pop	empty	20

Ans. Sequence of popped values, 20, 20, 10, 10, 20